

# Erik Henrique Alves Cunha

Senior Full-Stack Software Engineer | React, Next.js, Angular, Node.js, TypeScript, AWS

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## SUMMARY

I design frontend platforms, not frontend features: micro-frontend topologies, shared payment primitives, rendering and state architecture, async orchestration, performance, and the BFF and API layers underneath them. 8+ years building production frontend systems for fintech, healthcare, e-commerce, and banking. Primary stack is **React, Next.js, TypeScript, and Node.js**, with enterprise depth in Angular, RxJS, NgRx, Web Components, and Micro-frontends.

Systems shipped at meaningful scale: payment platforms processing **40M+ transactions a year and €1B+ in revenue**, commerce sites serving **8M+ monthly users**, real-time healthcare operations dashboards, and digital banking surfaces backing **70M+ customers**. Measurable results on Core Web Vitals, TTI, bundle size, deployment time, and conversion.

Currently building **AI engineering systems** inside the delivery loop: multi-agent workflows, spec-driven development, path-scoped tooling instructions, evaluation pipelines, and automated review, refactor, and scaffolding workflows, designed so every automated action stays reviewable.

## PROFESSIONAL EXPERIENCE

### Senior Software Engineer

06/2026 – Present

Raylu.ai, New York, US (Remote)

- Building the **frontend platform** for Raylu, an AI-native deal origination platform used by 45+ private-market investment funds representing \$500B+ in AUM: reusable infrastructure, shared UI systems, and state architecture for data-intensive investor workflows.
- Built a reusable headless **DataTable** feature on **TanStack Table**: a versioned **ViewState** contract, URL-synced deterministic state with backward-compatible codecs, and persistence separating immutable view identity from mutable runtime state, keeping business logic out of UI implementation. Covers hierarchical grouping, expansion state, pagination, filtering, sorting, and column management, with shareable, bookmarkable, refresh-safe views.
- Established accessibility standards across the platform's shared components by redesigning interaction models around **WAI-ARIA semantics**, keyboard navigation, accessible naming, live regions, row-header semantics, and screen-reader behavior, resolving multiple accessibility issues during development and code review.
- Established automated test coverage across serialization, state management, rendering behavior, accessibility, interaction flows, and regression scenarios, including Storybook interaction testing.
- Standardized engineering practices across the team through RFC-driven implementation, architectural decision records, contribution guidelines, and pull request conventions.
- Stack:** *React 19, Next.js 15, TypeScript, TanStack Table, TanStack Query, Tailwind CSS, Storybook, React Testing Library, Jest, Playwright, GitHub Actions, Vite.*

### Senior Software Engineer

03/2025 – 05/2026

Betsson Group, Malta (EU)

- Led frontend platform architecture and operational reliability for a PCI-DSS payment and cashier system that processed **40M+ transactions a year and generated €1B+ in revenue across 15+ regulated markets** for a publicly traded operator.
- The platform supported 8+ languages, 5+ brand variants, and 10+ currencies including zero-decimal. It was structured as a **micro-frontend topology over an Nx monorepo** so brand-specific orchestration stayed isolated from shared payment primitives.
- Defined the reusable payment primitives that ran across multiple production brands and host frameworks at the same time, using **StencilJS Web Components with JSX and React.js** runtime integration. One source of payment orchestration, many host runtimes.
- Designed the stateful multi-step deposit and withdrawal flows that had to satisfy audit and regulatory requirements across 15+ jurisdictions. Modeled the **async state machines and recovery paths in RxJS and NgRx** so error states were part of the design rather than tacked on later.
- Designed a 12-agent and subagent **AI-assisted engineering system** on GitHub Copilot covering code generation, review, debugging, testing, and architectural validation. Built as a delivery system rather than tooling, adopted team-wide: review cycle time dropped, throughput went up, and every automated action stayed reviewable by design.
- Established regression protection on Jest and Playwright, with WireMock isolating upstream APIs so high-risk KYC and payment flows were not exposed to drift in dependent services.
- Established the platform's frontend architecture standards: a 35+ page knowledge base of system boundaries, decision logs, and diagrams that codified architectural decisions and **cut onboarding time by about 40%** by replacing tribal knowledge with a written reference.
- Stack:** *TypeScript, React.js, JSX, StencilJS, Web Components, Micro-frontends, Nx Monorepo, Vite, Angular, RxJS, NgRx, Ember.js, HTML5, CSS3, SCSS, Storybook, REST APIs, Jest, Playwright, WireMock, GitHub Actions, GitHub Copilot.*

## Senior Software Engineering Consulting

04/2023 – 02/2025

Canon Medical Systems Brazil, *Canada (Remote)*

- Owned the **frontend platform end to end** for a mission-critical real-time healthcare operations system used across a multi-site regional hospital network. The product covers data-dense dashboards, scheduling, and live operational telemetry under high-frequency update load.
- Reduced the **JavaScript bundle by about 33% and the CSS bundle by about 98%** through lazy loading, code splitting, and a layout refactor. Initial load time dropped about 25%.
- Improved **Time to Interactive by about 52%** by removing waterfall request chains in favor of concurrent async loading, with TanStack Query cache reuse and component-level memoization (React.memo, useMemo) behind it. A rendering-path rework chosen over a caching change because the bottleneck was request sequencing, not data freshness.
- Automated infrastructure and releases with Bash, SSH, and NGINX. Manual production update time went from **about 5 minutes to about 45 seconds**, rollbacks down to about 1 minute, and environment-specific config changes about 50% faster.
- Built real-time operational dashboards on **WebSocket (STOMP/SockJS)** streams, with TanStack Query handling server state and isolating remote-data concerns from UI state. Reactive data flow feeds 4 complex dashboard views without UI thrashing under sustained event pressure.
- Established architectural boundaries separating domain, application, infrastructure, and presentation concerns to keep it maintainable as scope expanded. Lighthouse accessibility scores stayed near 100 across production views, with **WCAG 2.1 AA** compliance.
- Stack:** *React.js, TypeScript, TanStack Query, Nx Monorepo, Vite, WebSocket (STOMP/SockJS), REST APIs, PrimeReact, Chart.js, FullCalendar, Bryntum Scheduler, HTML5, CSS3, SCSS, Storybook, Jest, React Testing Library, Bash, SSH, NGINX, GitHub Actions, Lighthouse.*

## React Engineer @ Nike Brazil / Centauro

12/2021 – 04/2023

Grupo SBF (Nike Brazil, Centauro), *São Paulo, Brazil (Remote)*

- Built scalable frontend systems for high-traffic commerce platforms across Nike Brazil and Centauro, LATAM's largest sports retailer. The unified platform serves **8M+ monthly users across 2+ brand properties**, where LCP improvements show up directly in conversion.
- Delivered **SSR and SSG rendering with micro-frontend boundaries** across 2+ brand properties on the shared platform. Brands stay isolated without forking the codebase or coupling their release cycles.
- Reduced page load time by about 32% through code splitting, dynamic imports, image optimization, caching, and bundle analysis. **Core Web Vitals improved about 12%**, mostly from LCP.
- Lifted conversion by about 10% by setting up an experiment-driven workflow. Designed, shipped, and analyzed **20+ A/B experiments** through Google Optimize 360 and Google Analytics, with stop-loss criteria on each.
- Maintained Jest, React Testing Library, and MSW suites at around 70% coverage on 4+ revenue-critical user journeys. Mentored 4 junior engineers on React patterns, testing discipline, and review standards.
- Integrated a WebAR try-on into the purchase journey, which contributed to measurable uplift in add-to-cart and purchase completion within that flow.
- Stack:** *React.js, Next.js, TypeScript, SWR, styled-components, Nx Monorepo, Webpack, SSR/SSG, Micro-frontends, HTML5, CSS3, Storybook, REST APIs, Jest, React Testing Library, MSW, GitHub Actions, Google Optimize 360, Google Analytics.*

## Frontend Engineer @ VMware

01/2021 – 12/2021

Encora Inc. (Daitan Group), *Campinas, Brazil*

- Delivered enterprise frontend features for **VMware Pathfinder**, VMware's global product test-drive and hands-on labs platform, which has served **2.1M+ cumulative labs** across multi-language, multi-region users worldwide (vSphere+, vSAN+, Aria, Tanzu, Horizon Cloud, NSX, Cloud Foundation). Integrated Zoom APIs for embedded customer engagement workflows.
- Redesigned the **client-side state architecture** with Redux Toolkit to isolate concurrent asynchronous workflows across feature modules, organizing slices, selectors, and async thunks so concurrent requests no longer interfere with each other.
- Built automated regression pipelines reaching about 70% coverage on core business workflows. Contributed to CI/CD on AWS CodePipeline and CodeBuild, with static assets served from S3 and CloudFront.
- Stack:** *React.js, TypeScript, Redux Toolkit, React Hook Form, Webpack, HTML5, CSS3, SCSS, Jest, React Testing Library, Zoom API, AWS CodePipeline, CodeBuild, CodeDeploy, S3, CloudFront.*

## Frontend Engineer @ Itaú Bank

04/2020 – 01/2021

Zup Innovation (Itaú Bank), *Campinas, Brazil*

- Delivered frontend systems for digital banking platforms at Zup Innovation, the Itaú Unibanco-owned tech arm that serves Latin America's largest bank. The bank reaches **70M+ customers across 18+ countries** through regulated digital banking workflows.
- Led the modular UI architecture: reusable component libraries and Web Components enabling **micro-frontend integration** across multiple banking applications, decoupling release cycles between teams. Applied accessibility and compliance standards (semantic HTML, ARIA, keyboard navigation) across workflows that required regulatory validation.
- Expanded automated regression coverage across core transactional interfaces and shared components, raising baseline stability on compliance-sensitive flows.

- **Stack:** *TypeScript, Web Components, Micro-frontends, RxJS, Angular, Webpack, HTML5, SCSS, REST APIs, Karma, Jasmine, AWS CodePipeline, CodeBuild, CodeDeploy, S3, CloudFront.*

## Frontend Engineer

11/2019 – 04/2020

Venturus, Campinas, Brazil

- Built operational dashboards for CCR AutoBAN (CCR Group), operator of the **320 km Anhanguera-Bandeirantes (SP-330 + SP-348) toll corridor** connecting Greater São Paulo to Campinas and the country's interior. Modular React components and Redux Toolkit async data flows feed monitoring, filtering, and operational analysis views, with client-side CSV export and parsing so operations teams generate large reporting datasets directly from the dashboard.
- Added automated testing on complex data-driven UI components to improve regression protection across the dashboard surface.
- **Stack:** *React.js, TypeScript, Redux Toolkit, Webpack, HTML5, CSS3, SCSS, REST APIs, Jest, React Testing Library, Jenkins CI/CD.*

## Full-Stack Engineer (MEAN)

02/2019 – 11/2019

Venturus, Campinas, Brazil

- Led development of a foreign trade advisory platform for CGO Assessoria, a 30+ year Brazilian import/export and agribusiness customs specialist supporting compliance-driven workflows for international trade documentation.
- Resolved critical backend performance bottlenecks, reducing reporting API execution time by about **97.5% (from around 40s to under 1s)**, and other endpoints from around 2s to around 200ms. The work involved query redesign, indexing, and fixing ORM access patterns rather than caching shortcuts.
- Redesigned the core data architecture, migrating document-heavy workloads from MongoDB to PostgreSQL with relational modeling and indexing. Built a Node.js **ETL pipeline** for schema transformation, normalization, and validation across the migration. Introduced centralized observability with Sentry and standardized environments with Docker and Docker Compose, raising reproducibility across the engineering team.
- **Stack:** *Node.js, Express, TypeScript, PostgreSQL, MongoDB, Angular, RxJS, Docker, Docker Compose, Sentry, Jenkins.*

## Mobile Developer

09/2018 – 02/2019

MB Labs, Campinas, Brazil

- Architected SM Aprendizagem for Edições SM (Brazilian arm of Grupo SM, a Spanish educational publisher present in nearly 80 countries). It is an offline-first cross-platform app delivering didactic content across **5 operating systems** (Android, iOS, Windows, macOS, Linux) from a single codebase. Delivery cost dropped about 80% versus separate native codebases, with around 90% logic reuse.
- Designed an **offline-first sync engine** over REST APIs with incremental updates and conflict resolution. Recorded engagement events locally in SQLite and synced on reconnection with zero data loss across sessions and device types.
- **Stack:** *Ionic, Angular, TypeScript, Electron, SQLite, REST APIs, HTML, CSS, SCSS.*

## Full-Stack Developer

07/2018 – 09/2018

MB Labs, Campinas, Brazil

- Built the React frontend and Node.js backend services for the Hondana INDICA enterprise chatbot platform, a B2B HR-tech product integrated with Facebook Workplace and used by clients including Cia Hering (which reported 450% greater engagement versus prior training methods) and KPMG Business School. Designed **asynchronous webhook-based event pipelines** with explicit retry logic and backpressure handling, plus batch push notification delivery on Firebase Cloud Messaging.
- **Stack:** *React.js, Node.js, Express, JavaScript, Webpack, HTML5, CSS3, SCSS, REST APIs, Webhooks, Firebase Cloud Messaging, MS SQL Server, AWS (EC2, RDS, S3, Route 53), NGINX.*

## Frontend Engineer

10/2017 – 05/2018

Monde Sistemas, Americana, Brazil

- Built reusable Vue.js components, SPA views, and shared design-system patterns for Monde Sistemas, a B2B SaaS travel-agency management (CRM/ERP) platform used by **2,200+ Brazilian travel agencies**, integrating RESTful APIs on the JSON:API specification and adding automated unit testing with Jest and Sinon across shared frontend modules.
- **Stack:** *Vue.js, JavaScript (ES6+), Webpack, HTML5, CSS3, Stylus, REST APIs, JSON:API, Jest, Sinon, GitHub.*

## SELECTED PROJECTS

**erikunha.dev** (Author, [github.com/erikunha/portfolio](https://github.com/erikunha/portfolio)): Single-page Next.js 15 + React 19 portfolio on Vercel Edge. RSC-first composition with intentional client islands, a streaming Anthropic LLM endpoint with prompt caching and a monthly token budget, durable contact form with KV-before-delivery semantics, and a CI pipeline that treats performance, accessibility, and bundle size as hard contracts (LCP  $\leq 1.8s$ , INP  $\leq 200ms$ , JS per-route  $\leq 120KB$  gzipped, Lighthouse a11y = 100). TypeScript strict, Biome, Vitest, Playwright + axe-core, Lighthouse CI. Built under a written AI-assisted development contract (CLAUDE.md) with constraints enforced by tooling rather than discipline.

**LAibemage** (Author, npm package): Open-source TypeScript CLI and SDK that classifies, timestamps, and labels images through LLM Vision APIs across 7 providers (OpenAI, Anthropic, Google, Azure, Ollama, AWS Bedrock, Vertex AI), with a hybrid local-to-cloud routing tier. Features async batch processing (~50% cost reduction on supported providers), perceptual-hash burst dedup, temporal consensus voting on ambiguous classifications, embedding-based semantic search, and a plugin lifecycle. Pure TypeScript with Vitest, full CI on GitHub Actions, Zod-validated config, and a published npm SDK.

OPEN SOURCE

[react-google-maps-api](#) (OSS contributor, PR #3266): Diagnosed and fixed a React Hooks infinite re-render loop in the Directions API example of react-google-maps-api, a popular community library with ~700K weekly npm downloads. Isolated the root cause (a callback prop triggering setState on every fire, recreating the callback identity, looping forever), reproduced it on a minimal StackBlitz, and submitted a PR with a stable-callback fix, after personally incurring the cost of 61K+ unwanted Google Maps API calls from the library's official docs site. Coordinated diagnosis with the maintainer and other contributors against a competing PR that did not resolve the loop.

SKILLS

**Frontend Architecture and Platform:** React.js, Next.js (App Router, RSC, Server Actions), TypeScript, JavaScript (ES6+), Angular, Micro-frontends, Web Components (StencilJS), Nx monorepo, component-driven architecture, design systems, BFF and API integration layers, state isolation, Vite, Webpack, Storybook, JSX, HTML5, CSS3, SCSS, styled-components, Tailwind

**Rendering, Performance, and Accessibility:** SSR/SSG, Core Web Vitals (LCP, INP, CLS), TTI optimization, bundle analysis, code splitting, image optimization, WCAG 2.1 AA, ARIA

**State and Async Orchestration:** Redux Toolkit, NgRx, RxJS, TanStack Query, SWR, React Hook Form, async state machines, event-driven data flows

**Backend and Data:** Node.js, Express, REST APIs, Webhooks, event-driven pipelines, PostgreSQL, MongoDB, MS SQL Server, SQLite, ETL, query optimization

**Reliability and Testing:** Jest, Playwright, React Testing Library, MSW, WireMock, Karma, Jasmine, regression coverage on revenue-critical journeys

**Cloud and Delivery:** AWS (CodePipeline, CodeBuild, S3, CloudFront, EC2, RDS, Route 53), Docker, NGINX, Jenkins, GitHub Actions, Sentry

**AI Engineering Systems:** Multi-agent and subagent configurations, spec-driven development, evaluation pipelines, path-scoped engineering instructions, automated review, refactor, and scaffolding workflows, GitHub Copilot, Claude

EDUCATION

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|-------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Web Development Specialist Co-op Diploma</b><br>Cornerstone International Community College of Canada (CICCC), Vancouver, Canada | 08/2023 – 02/2024 |
| <b>Bachelor's of Information Systems</b><br>Centro Universitário Salesiano de São Paulo (UNISAL), Campinas, Brazil                  | 01/2015 – 06/2020 |

CERTIFICATIONS, LANGUAGES AND COMMUNITY

**Certifications:** WES Verified International Academic Qualifications (2022). IELTS General Training C1 (2023). Angular Developer Certification, Alain Chautard / Google Developer Expert (2024).

**Languages:** Portuguese (Native), English (C1), Spanish (B1), French (A2).

**Volunteering, DevOpsDays Campinas (Organizer, 2024):** Led speaker sourcing, evaluation, and selection across DevOps, cloud infrastructure, platform engineering, and software architecture submissions. Curated a 10+ talk technical program and coordinated speakers, sponsors, and co-organizers across the full event cycle.